

HASHMAP

HashMap have constant searching time

$O(1)$

HashMap
 → why
 → How it works.

Structure of HashMap

Key (Name)	Value (marks)
Kunal	68
Saaim	32
apoorv	88

if we use `map.get()` like `map.get("saaim");` it will be in constant time like $O(1)$ no time needed. ~~That~~ why we use it.

Problems we face during hashing?

- ① we need hashcodes for the storing of elements
- ② we need to reduce those hashcodes.

like `hash("saaim") = 7`

we do it by hashcode function `hashCode();`

9

So hashmap stores those on the basis of hashcode that's why its traversal is so fast.

methods:

map.putIfAbsent (key, value);

map.replace (key, value) or (key)

map.clear() // To remove all

map.put (key, value);

map.remove (key) or (key, value)

map.compute or map.merge

↘ ↙
high level methods.

Also

if we return map.values()
it return in form of
Collection class like array list
[...].

not keys only values.